Deriving the Constant Rate Effect in Language Change

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Part I

Introduction

Languages and grammars

- A **language** is an infinite set of strings over some alphabet
- A **grammar** G for a language L is a finite object that generates all and only the strings in L
- Speakers' knowledge of their language is encoded in the form of such grammars, which are to be taken as abstract mental representations
- Learning theory attempts to explain how children acquire G based on a finite amount of input ('primary linguistic data', or PLD)

Languages and grammars

- During periods of stability, all members of a language community share the same *G*
- Languages, however, do change
  - Old English allowed O(bject)V(erb) word order; Modern English is VO:

    (1) þæt he his stefne up ahof
    that he his voice up raised
    'that he raised his voice'

    (2) * that he his voice raised

  - Scottish English merges two traditionally distinct sound categories, /ɔ/ and /ɒ/:

    (3) I caught /kɔt/ a fish

    (4) My cot /kɒt/ > /kɔt/ is comfortable

Languages and grammars

- Unlike Old and Middle English, Modern English has obligatory periphrastic *do* in questions and negative declaratives:

  (5) How great and greuous tribulations suffered the Appostyls . . . ?
  how great and grievous tribulations suffered the Apostles

  (6) How great and grievous tribulations did the Apostles suffer?

  (7) . . . sondrie fraudes, whiche he perceiueth not
  sondry frauds, which he perceives not

  (8) sundry frauds, which he does not perceive

The Constant Rate Effect

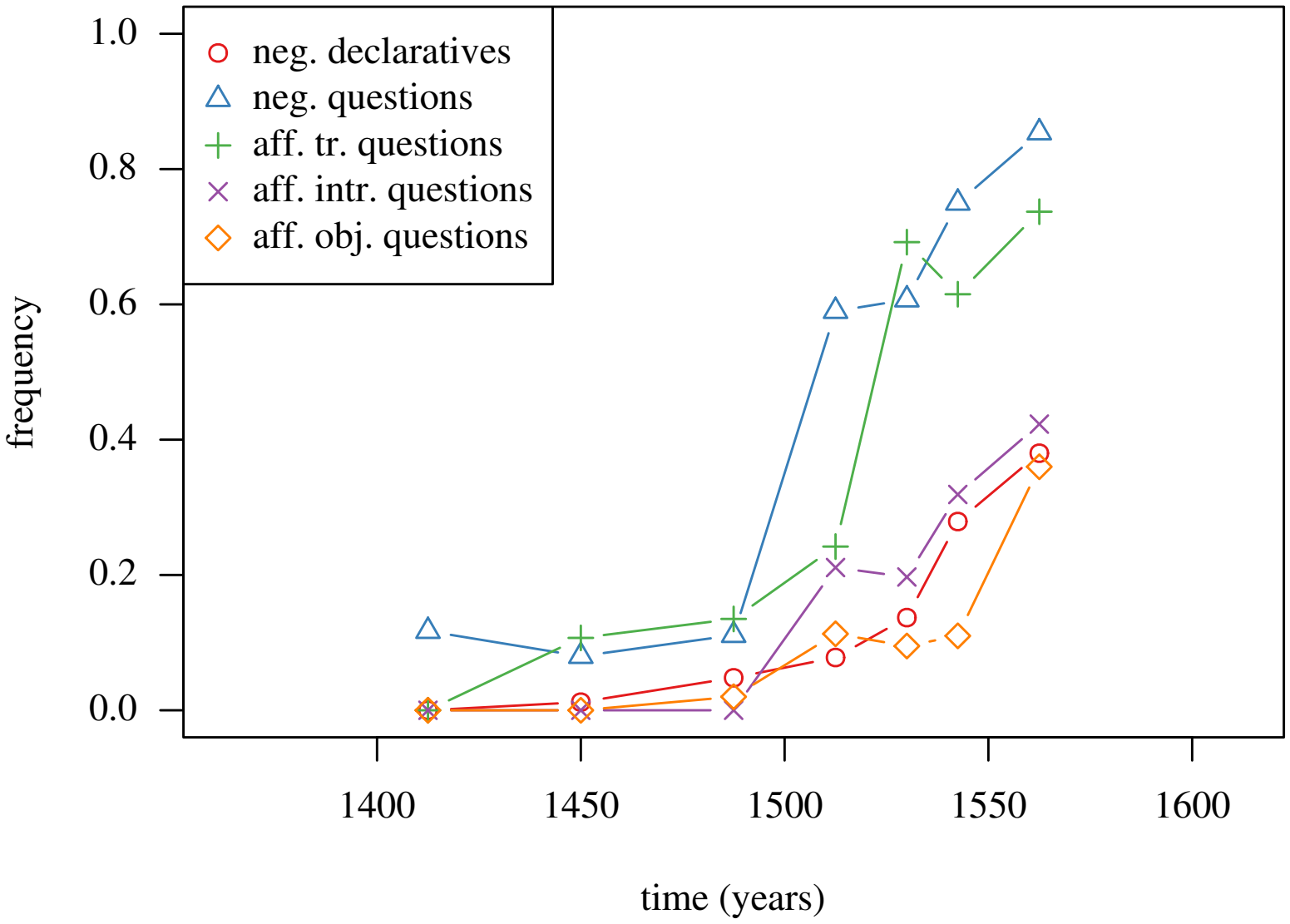
- Features of a grammar may surface differently in different linguistic contexts
  - E.g. in Early Modern English, periphrastic *do* is more frequent in questions than in negative declaratives (Kroch, 1989)

Constant Rate Hypothesis (Kroch, 1989, 200)

'[W]hen one grammatical option replaces another with which it is in competition across a set of linguistic contexts, the rate of replacement, properly measured, is the same in all of them.'

- This hypothesis has received enough empirical support over the last two decades for it to be referred to as the **Constant Rate Effect**, or **CRE** (e.g. Pintzuk, 1995)

The Constant Rate Effect



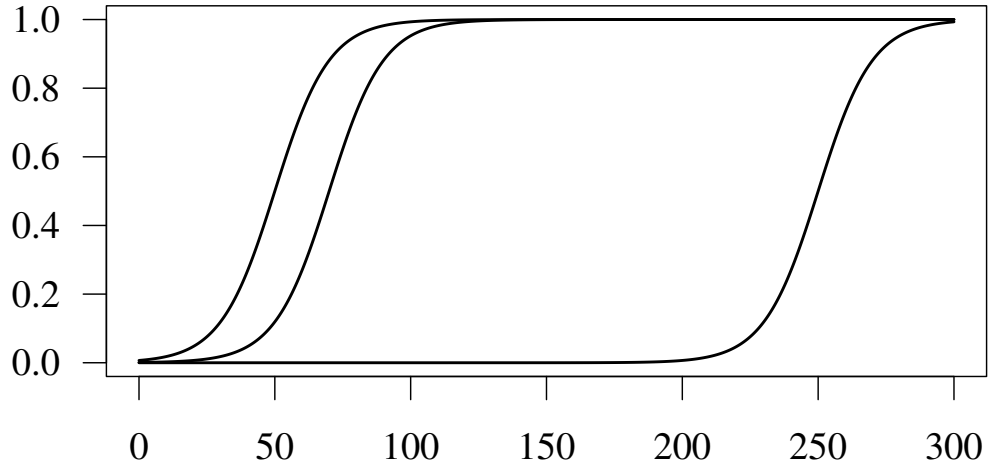
The Constant Rate Effect

- Language change commonly modelled using the logistic

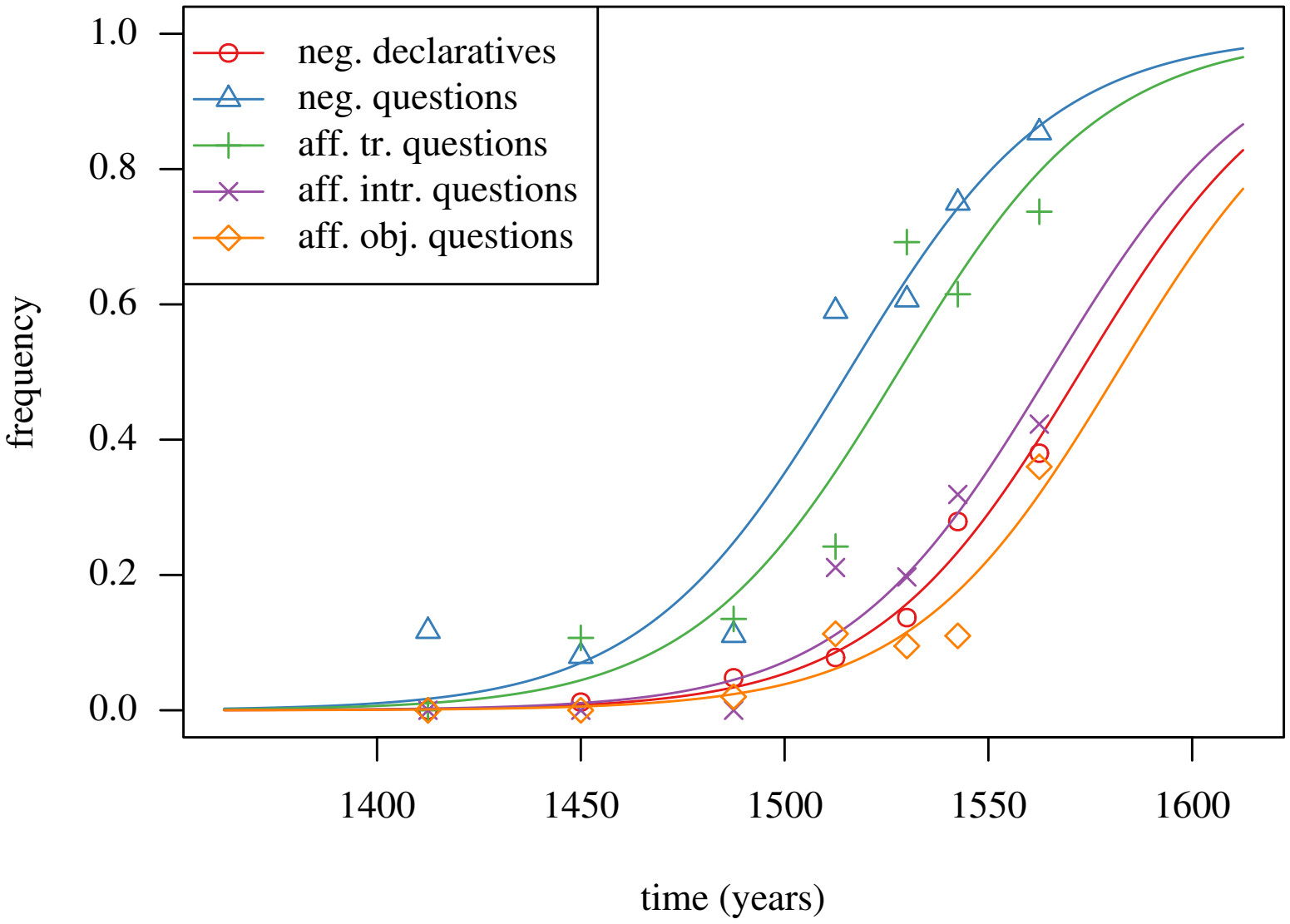
$$p_t = \frac{1}{1 + e^{-s(t-k)}}, \tag{1}$$

one per each context of interest

  - Rate of change s often called **slope** in the literature, and k the **intercept**
  - CRE diagnosed if, for two or more contexts, the slopes s are more or less identical



The Constant Rate Effect



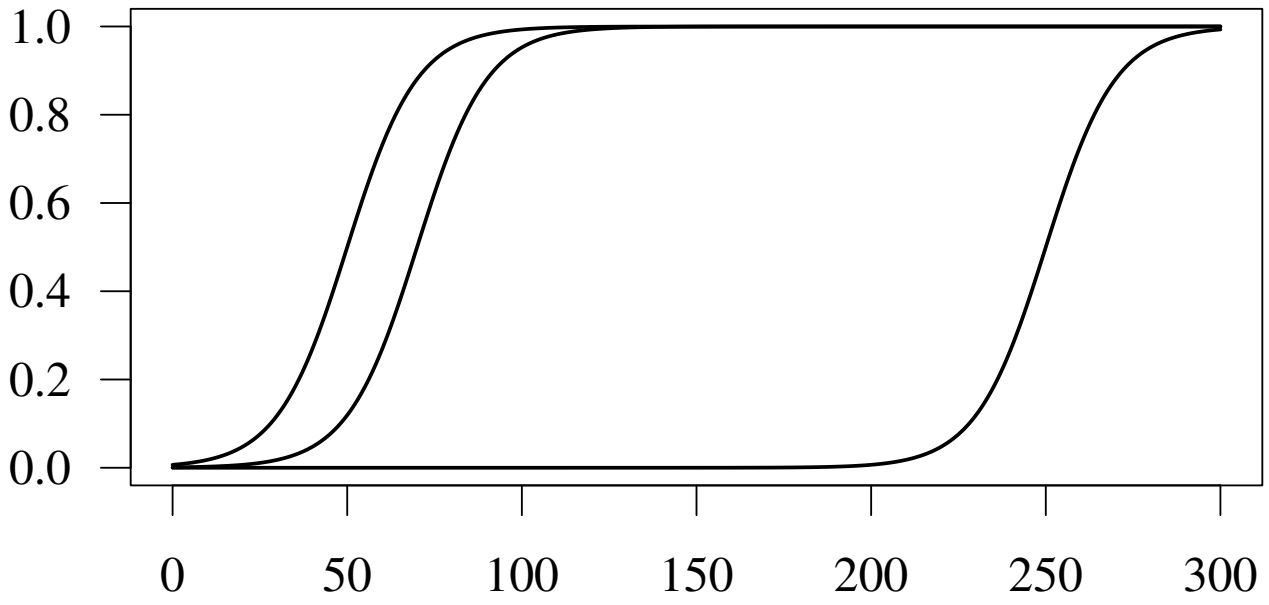
The Constant Rate Effect

- Key intuition behind the Constant Rate Hypothesis:
  - A single grammatical parameter governs all contexts ...
  - ...but some contexts may favour the new grammar ...
  - ...while others disfavour it, leading to differences in observed frequencies
- These (dis)favouring effects (**contextual effects**) are constant across time and independent of the underlying grammatical change
- Key assumption: speakers have access to multiple competing grammars and use them with different frequencies

Problems

- Classical CRE literature fits independent logistics to a number of linguistic contexts, guided by the "identical slopes, different intercepts" intuition
- It does not offer a **mechanism** for the CRE
- This implies two problems:
  - Variation in k is entirely unconstrained
  - Slope parameters s may be same for reasons other than underlying grammatical unity

Problems, 1: Variation in k



Problems, 2: Identity of *s*

- Wallenberg (2015): relative clause extraposition is declining in the histories of both English and Icelandic, with similar *s* parameters
- Willis (2015): for the rise of the second-person pronoun *chdi* in Welsh, the *s* is same across different geographical regions of Wales
- Corley (2014): for negative concord in Early Modern English, *s* are similar for male and female speakers

Problems

- Thus, the traditional "identical slopes, different intercepts" operationalization of the CRE is susceptible to false positives
- It also gives no causal mechanism for the CRE
  - Why should slopes s be same across contexts?
  - How could two **independent** logistics describe two contexts of **one and the same** underlying change (two Verhulstian processes vs. one such process)?

In this talk

- The aim of this talk is to introduce a mechanistic model of the CRE that does not suffer from the above-mentioned problems
- Starting with Yang's (2000) mathematical model of grammar competition through variational learning, we augment the model with production biases to account for the CRE
- This leads to a result, the Time Separation Theorem, which solves Problem 1 above
- It also leads to an increase in our understanding of how different factors interact to produce language change
  - Specifically, we show that Yang's (2000) Fundamental Theorem of Language Change fails to hold in the extended model, and demonstrate the existence of a more complicated bifurcation scenario
- Additionally, we test the model against three CREs and one pseudo-CRE

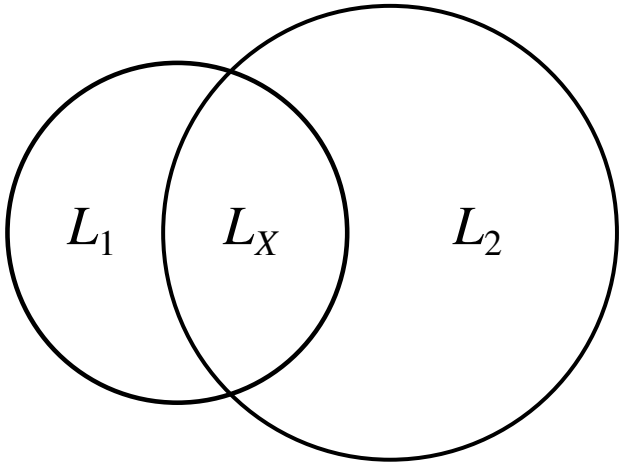
Part II

Model

Competing grammars (Yang, 2000)

- Assume grammars G_1 and G_2 compete, so that an individual uses G_1 with probability p_t and G_2 with probability $q_t = 1 - p_t$.

- The **advantage** of G_1 (resp. G_2), denoted α (β), is the proportion of sentences G_2 (G_1) cannot parse.



- Learner receives N input sentences from distribution

$$\begin{array}{ccccc} & L_1 & L_2 & L_X \\ G_1 & \alpha p_t & 0 & (1-\alpha)p_t \\ G_2 & 0 & \beta q_t & (1-\beta)q_t \end{array} \tag{2}$$

Competing grammars (Yang, 2000)

- Learner sets q_{t+1} (and $p_{t+1} = 1 - q_{t+1}$) in accordance with linear reward–penalty learning (Bush and Mosteller, 1951)
- Initially, set $p^{(0)} = q^{(0)} = 0.5$. For input sentence $s = 1, \ldots, N$:
  - if G_1 parses sth sentence, then

$$\begin{cases} p^{(s)} = p^{(s-1)} + \gamma(1 - p^{(s-1)}) \\ q^{(s)} = (1-\gamma)q^{(s-1)} \end{cases} \tag{3}$$

  - if G_1 does not parse sth sentence, then

$$\begin{cases} p^{(s)} = (1-\gamma)p^{(s-1)} \\ q^{(s)} = \gamma + (1-\gamma)q^{(s-1)} \end{cases} \tag{4}$$

  for some fixed learning rate $0 < \gamma < 1$.
- Set $p_{t+1} = p^{(N)}$ and $q_{t+1} = q^{(N)}$

Competing grammars (Yang, 2000)

- ICBS: as $N \to \infty$,

$$q_{t+1} \to \frac{\beta q_t}{\beta q_t + \alpha p_t} = \left(1 + \rho \frac{1 - q_t}{q_t}\right)^{-1}, \tag{5}$$

where $\rho = \alpha / \beta$.

Theorem (The Fundamental Theorem of Language Change; Yang, 2000)

Let $\rho < 1$ and $N \to \infty$. Then $q_t \to 1$ as $t \to \infty$.

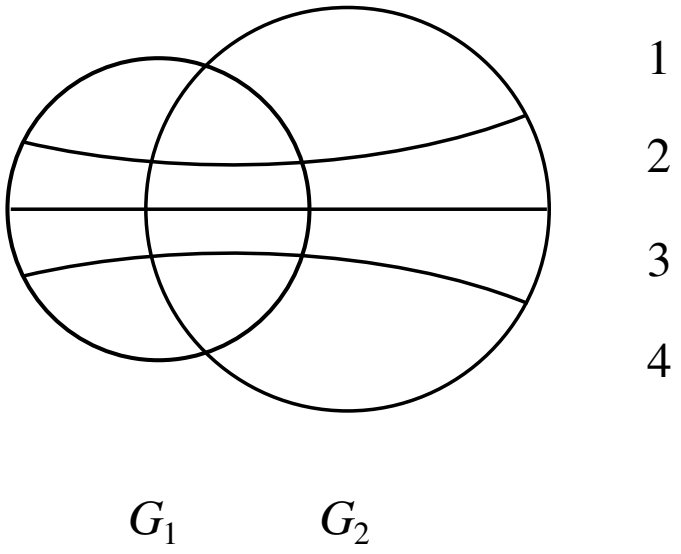
- ICBS: evolution of q_t is logistic with growth rate $s = -\log(\rho)$ and intercept $k = -\frac{1}{\log(\rho)} \log\left(\frac{1-q_0}{q_0}\right)$.

Competing grammars (Yang, 2000)

- Note assumptions:
  - Infinite well-mixing population with non-overlapping generations (or, equivalently, a single learner per generation)
  - Large N – learners are near-perfect probability matchers
  - Just two competing grammars

Production biases

- Now assume K contexts $1, \ldots, K$ which partition $L_1 \cup L_2 \cup L_X$
- Assume the ith context carries a proportion λ_i of all sentences



- Key idea: the CRE results from the fact that **production biases** of differing magnitudes affect different contexts

Production biases

- Let $q_t^{(i)}$ denote the probability with which speaker uses G_2 in the ith context (and similarly $p_t^{(i)}$ for G_1)
- Then the most general form of biasing is

$$\begin{cases} p_t^{(i)} = p_t + F(b_i, p_t, q_t) \\ q_t^{(i)} = q_t + G(b_i, p_t, q_t) \end{cases} \tag{6}$$

for some functions F, G, where b_i is the strength of bias in the ith context, with two requirements:

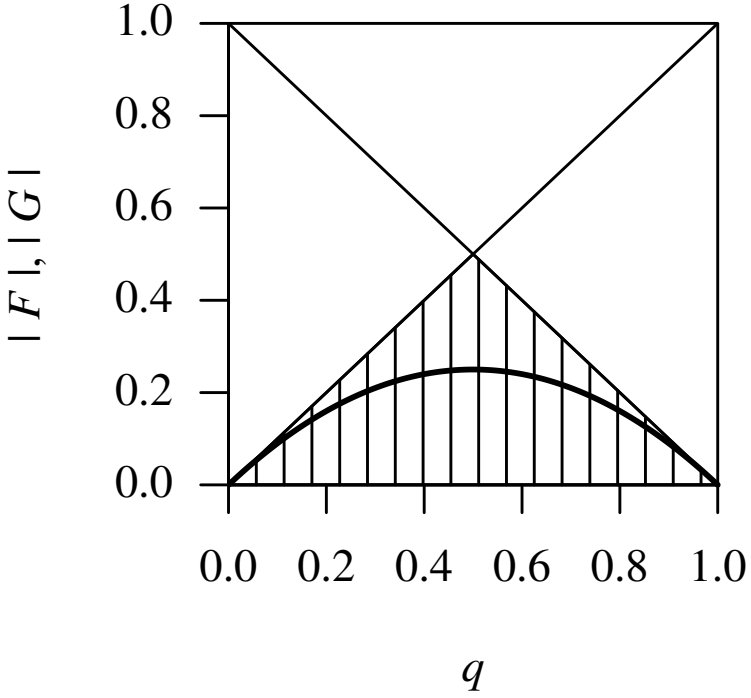
$$\begin{cases} p_t^{(i)} + q_t^{(i)} = 1 & \text{(as } G_1 \text{ and } G_2 \text{ are the only grammars)} \\ 0 \leq p_t^{(i)}, q_t^{(i)} \leq 1 & \text{(as the two quantities are probabilities)} \end{cases} \tag{7}$$

Production biases

Theorem

F and G satisfy (6) *and* (7) *if and only if*

$$\begin{cases} F = -G \\ |F|, |G| \leq \min\{p_t, q_t\} \end{cases}.$$



- Most parsimonious choice: $F = -b_i p_t q_t$ and $G = b_i p_t q_t$ with $-1 \leq b_i \leq 1$.
- Thus, probability to encounter a sentence from the ith context generated…

$$\ldots \text{by } G_1 = \alpha\lambda_i(p_t - b_i p_t q_t) \qquad \ldots \text{by } G_2 = \beta\lambda_i(q_t + b_i p_t q_t)$$

Production biases

- The Yangian equation

$$q_{t+1} = \left(1 + \rho \frac{1 - q_t}{q_t}\right)^{-1} = \left(1 + \frac{\alpha p_t}{\beta q_t}\right)^{-1} \tag{8}$$

now becomes

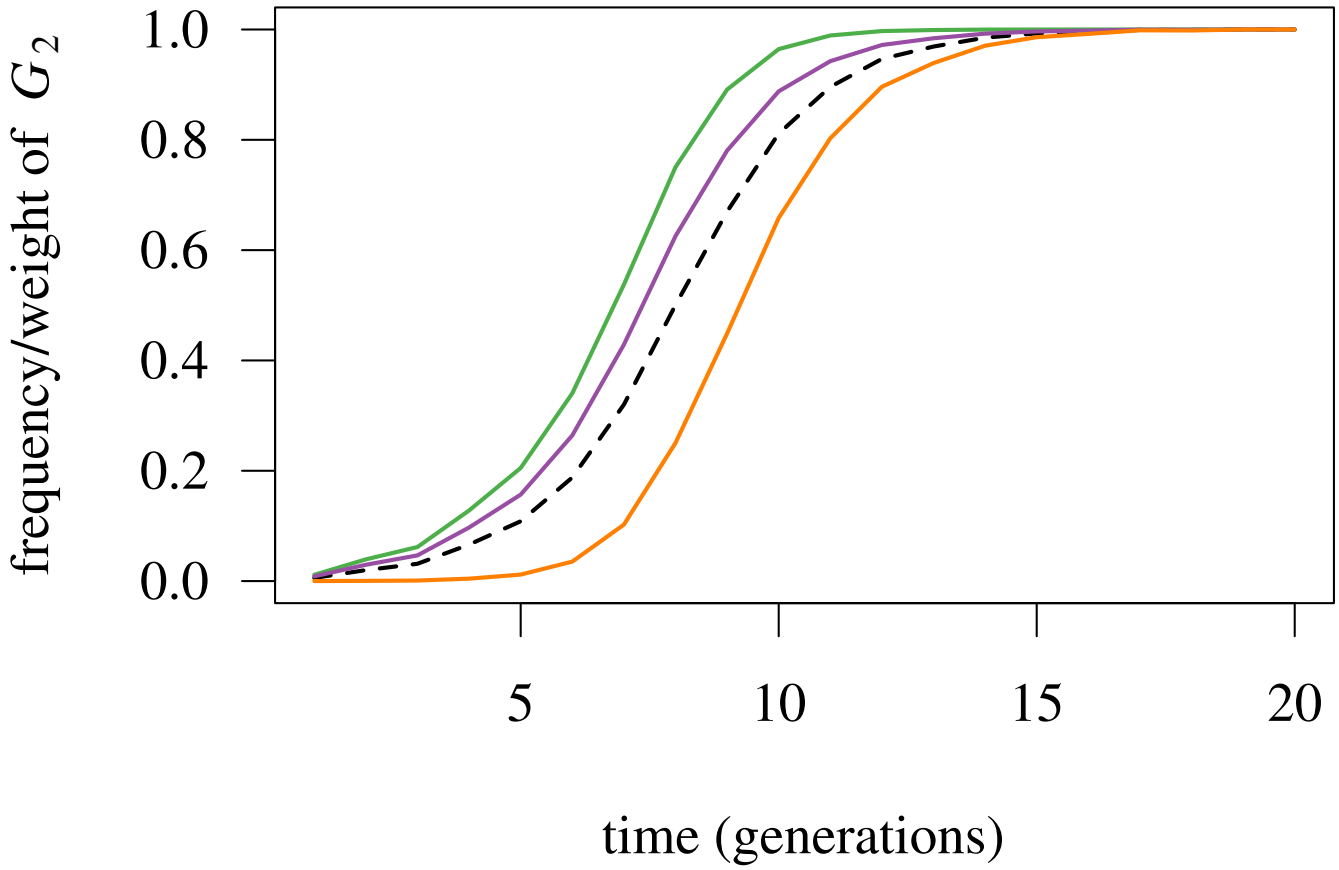
$$q_{t+1} = \left(1 + \frac{\sum_{i=1}^{K} \alpha \lambda_i (p_t - b_i p_t q_t)}{\sum_{i=1}^{K} \beta \lambda_i (q_t + b_i p_t q_t)}\right)^{-1} = \left(1 + \Lambda_t \rho \frac{1 - q_t}{q_t}\right)^{-1} \tag{9}$$

where $\rho = \alpha / \beta$ and

$$\Lambda_t = \frac{1 - q_t \sum_{i=1}^{K} \lambda_i b_i}{1 + (1 - q_t) \sum_{i=1}^{K} \lambda_i b_i} = \frac{1 - q_t B}{1 + (1 - q_t) B}. \tag{10}$$

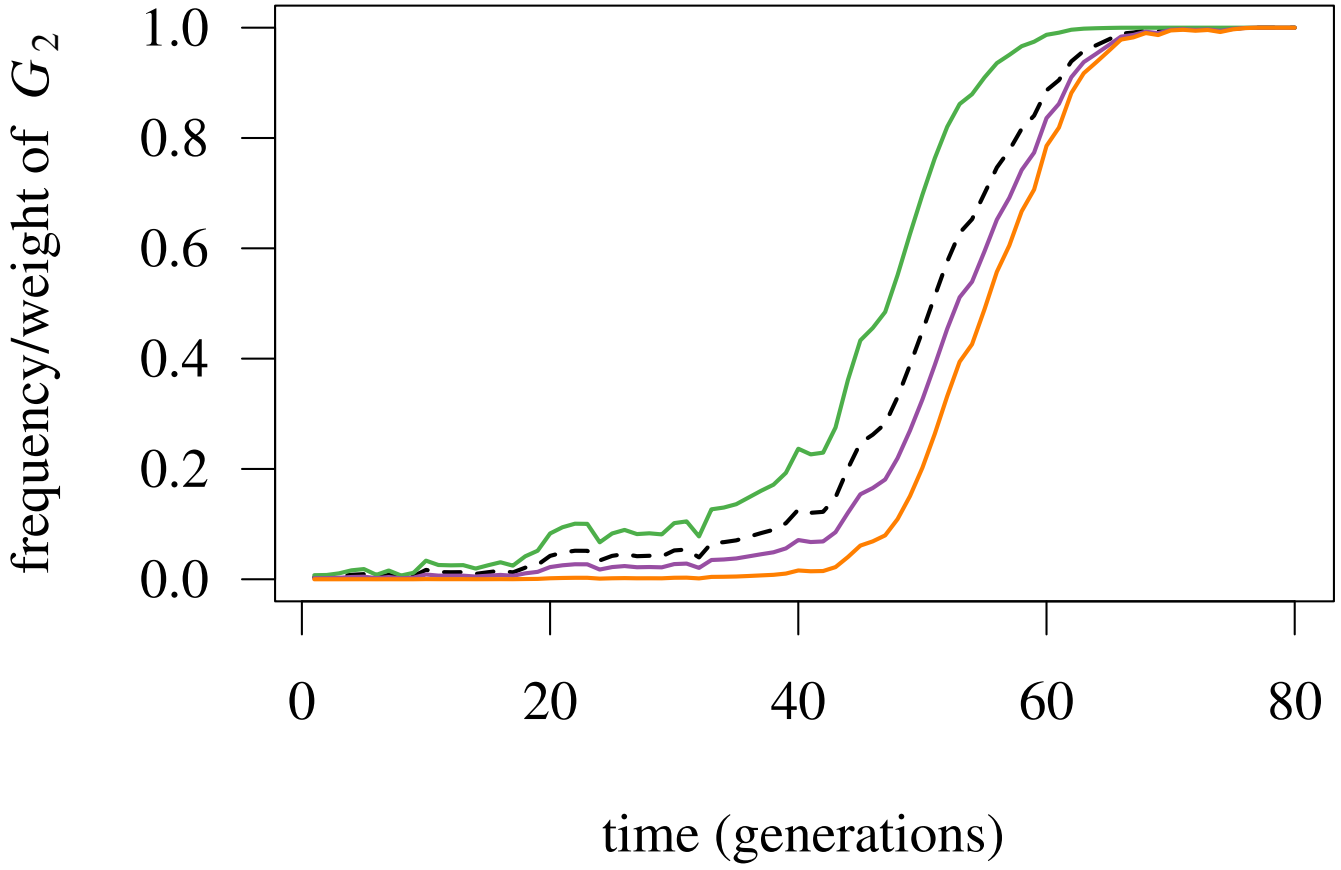
- Solution not known, so will have to approximate in the following

The Constant Rate Effect



$$N = 20000,\ \rho = 0.5,\ B = \sum \lambda_i b_i = 0$$

The Constant Rate Effect



$$N = 20000,\ \rho = 0.5,\ B = \sum \lambda_i b_i = -0.45$$

Outcome of grammar competition

- Observation: for suitable combinations of parameter values, change from G_1 to G_2 may be blocked even if $\beta > \alpha$ ($\rho < 1$)
- And: combinations of parameter values exist so that G_2 overtakes G_1 even though $\beta < \alpha$
- The quantity $B = \sum \lambda_i b_i$, **bias balance**, is crucial here

Outcome of grammar competition

Theorem (The Extended Fundamental Theorem of Language Change)

Let q_0 be the weight of grammar G_2 at point of actuation, let $B = \sum_{i=1}^{K} \lambda_i b_i$, and let

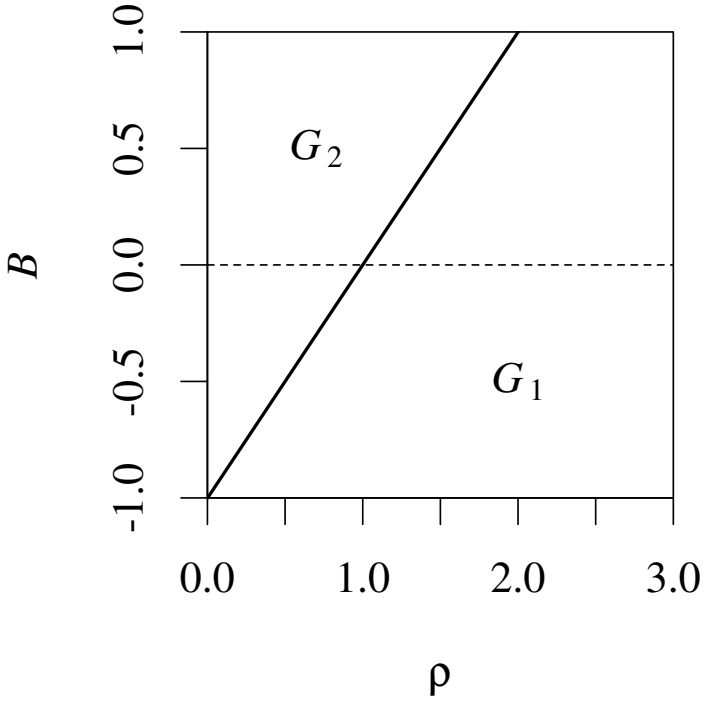
$$B_c = \frac{\rho - 1}{1 + q_0(\rho - 1)}.$$

Then in the limit $N \to \infty$,

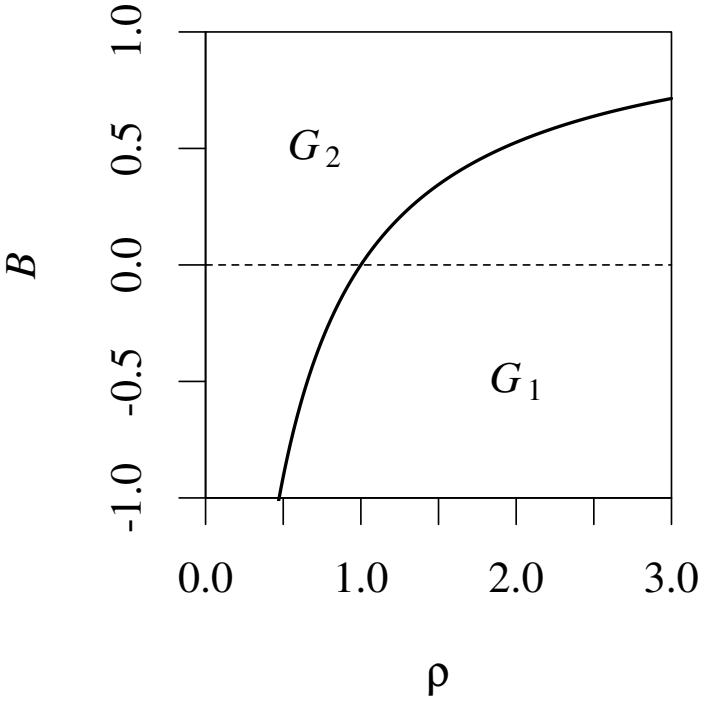
1. $q_t \to 1$ as $t \to \infty$, if $B > B_c$;
2. $q_t = q_0$ for all t, if $B = B_c$;
3. $q_t \to 0$ as $t \to \infty$, if $B < B_c$.

In other words, G_2 overtakes G_1 if, and only if, $B > B_c$.

Outcome of grammar competition



$q_0 = 0.001$



$q_0 = 0.9$

Logistic approximation

- Don't know how to solve crucial difference equation

$$q_{t+1} = \left(1 + \Lambda_t \rho \frac{1 - q_t}{q_t}\right)^{-1} \tag{11}$$

where

$$\Lambda_t = \frac{1 - q_t \sum_{i=1}^{K} \lambda_i b_i}{1 + (1 - q_t) \sum_{i=1}^{K} \lambda_i b_i} = \frac{1 - q_t B}{1 + (1 - q_t) B}. \tag{12}$$

- Strategy: reduce to a simpler formulation
  - Full model allows K arbitrary contexts
  - However, qualitatively, contexts are of three kinds: G_2-favouring, G_2-disfavouring, or neutral

Logistic approximation

- Let
  - $\lambda_1 = \nu$ be the proportion of neutral sentences
  - $\lambda_2 = (1-\nu)\varphi$ be the proportion of G_2-favouring sentences
  - $\lambda_3 = (1-\nu)(1-\varphi)$ be the proportion of G_2-disfavouring sentences
- Then Λ_t simplifies to

$$\begin{aligned}\Lambda_t &= \frac{1 - q_t\left(\nu \cdot 0 + (1-\nu)\varphi b + (1-\nu)(1-\varphi)(-b)\right)}{1 + (1-q_t)\left(\nu \cdot 0 + (1-\nu)\varphi b + (1-\nu)(1-\varphi)(-b)\right)} \\ &= \frac{1 - q_t(1-\nu)(2\varphi - 1)b}{1 + (1-q_t)(1-\nu)(2\varphi-1)b}.\end{aligned} \tag{13}$$

- Remove time dependence by assuming $q_t = 0.5$ for all t:

$$\tilde{\Lambda} = \frac{1 - \frac{1}{2}(1-\nu)(2\varphi-1)b}{1 + \frac{1}{2}(1-\nu)(2\varphi-1)b}. \tag{14}$$

Logistic approximation

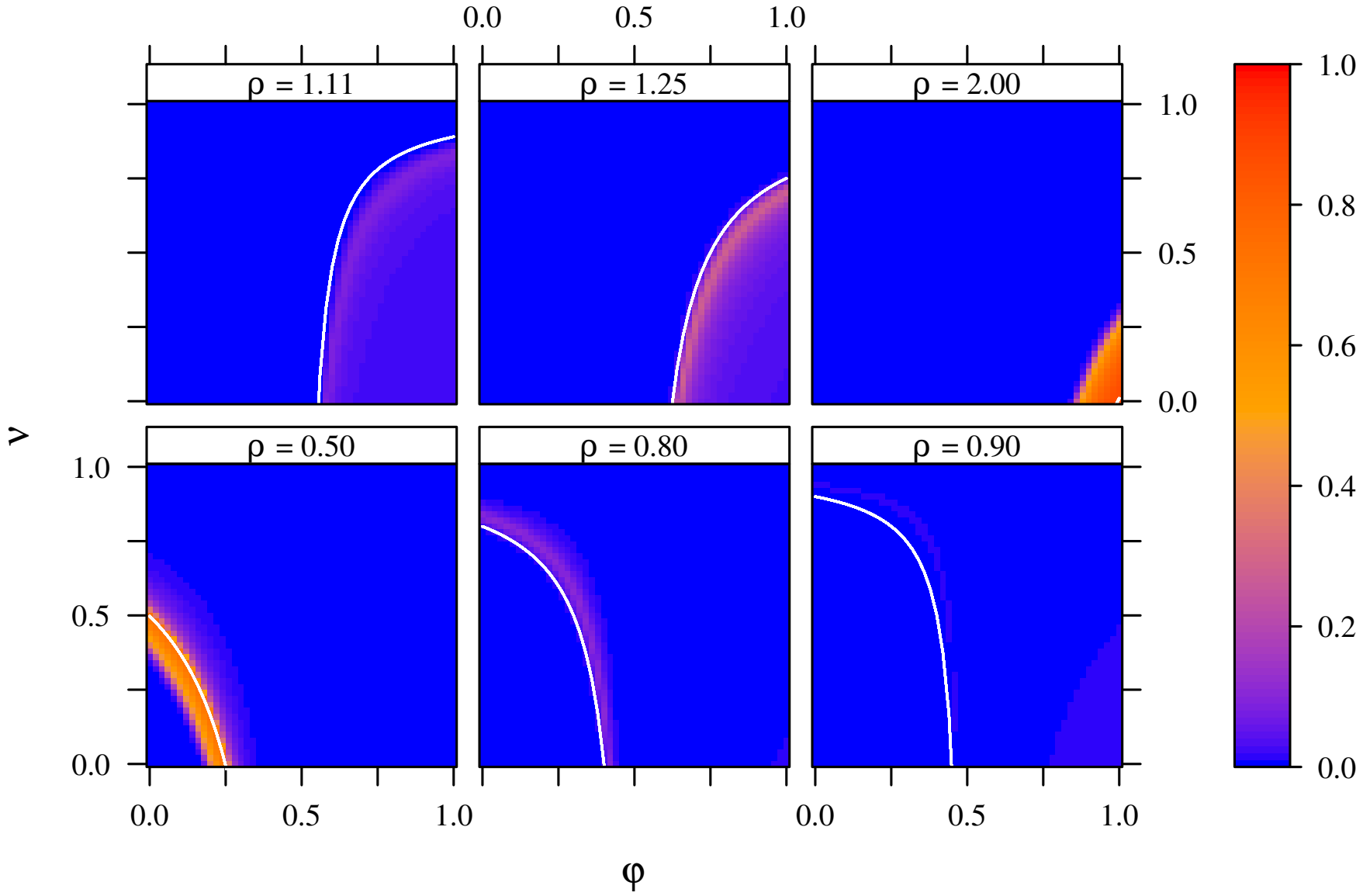
- Now

$$\tilde{q}_{t+1} = \left(1 + \tilde{\Lambda}\rho \frac{1 - \tilde{q}_t}{\tilde{q}_t}\right)^{-1} \tag{15}$$

is easily solved to yield the logistic

$$\begin{aligned}\tilde{q}_t &= \left(1 + (\tilde{\Lambda}\rho)^t \frac{1 - q_0}{q_0}\right)^{-1} \\ &= \left(1 + \exp\left(\log(\tilde{\Lambda}\rho)\left(t + \frac{1}{\log(\tilde{\Lambda}\rho)} \log\left(\frac{1 - q_0}{q_0}\right)\right)\right)\right)^{-1}\end{aligned} \tag{16}$$

- Is the approximation good?


ρ = 1.11
ρ = 1.25
ρ = 2.00
ρ = 0.50
ρ = 0.80
ρ = 0.90
ν
φ
0.0
0.5
1.0
0.2
0.4
0.6
0.8

Part III

Evaluation

Model evaluation

- With the logistic approximation, our aim is to fit to data a set of curves
$$q_t^{(i)} = \tilde{q}_t + b_i \tilde{q}_t (1 - \tilde{q}_t), \tag{17}$$
one per context, where $\tilde{q}_t$ is logistic as above.
- This approximation gives us, *inter alia*, the following useful result:

Theorem (The Time Separation Theorem)

For any two contexts of an underlying change proceeding at rate $s = -\log(\tilde{\Lambda}\rho)$, the maximal separation at the tipping points (where frequency $= 0.5$) is

$$\Delta(s) = \frac{2}{|s|} \log\left(\frac{1}{\sqrt{2} - 1}\right).$$

Model evaluation

- We now proceed to fit the model to a number of case studies: changes for which a CRE has previously been reported in the literature
  - emergence of periphrastic *do* in Early Modern English
  - Jespersen's Cycle in Middle English
  - loss of final fortition in Early New High German
- The idea is to assess the goodness of fit by traditional means, but also to see whether the data fit the predictions of our Time Separation Theorem

Periphrastic *do*

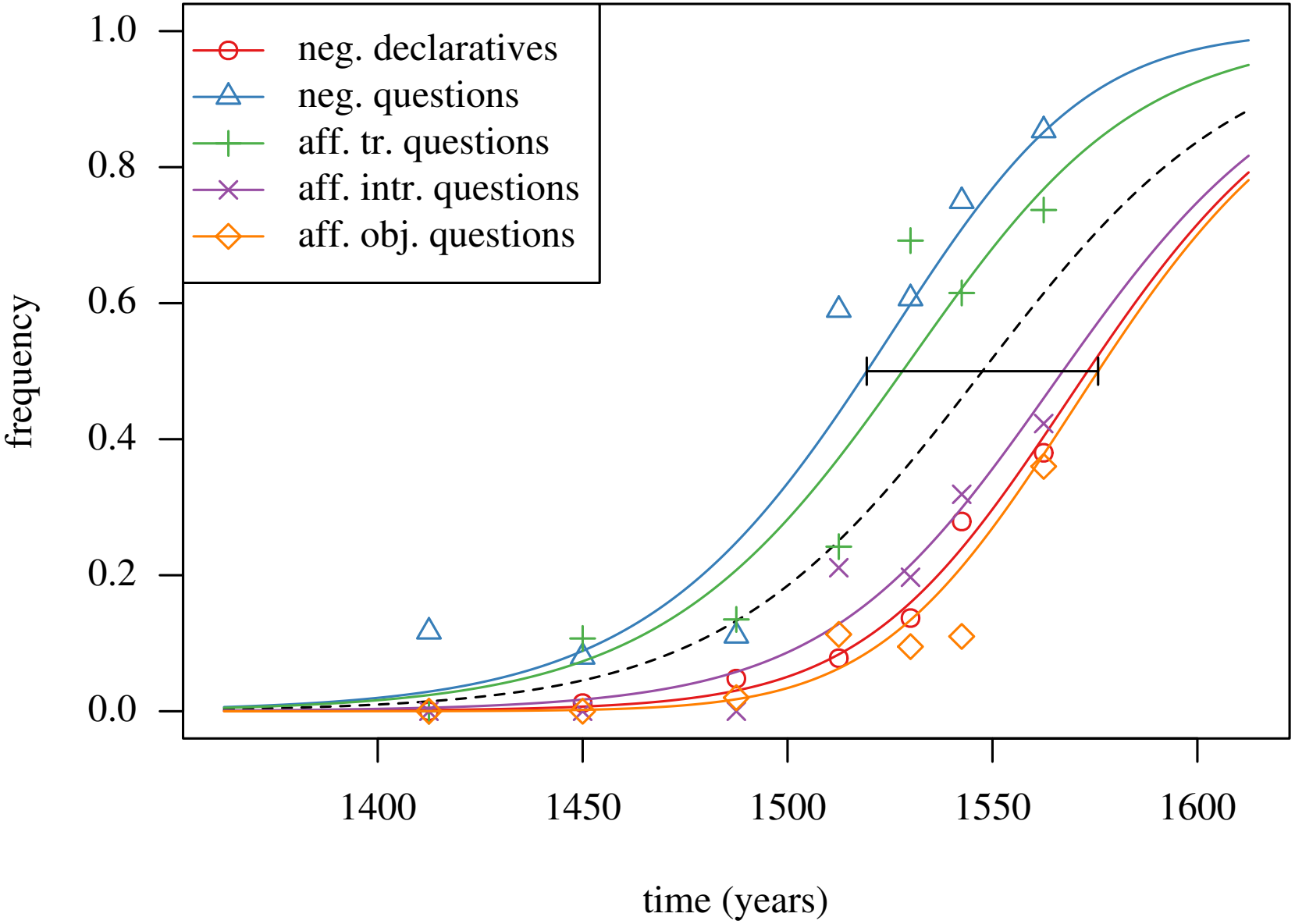
- Modern English uses periphrastic *do* in certain contexts, such as all interrogatives and negative declaratives

  (9) How great and greuous tribulations suffered the Appostyls?
  how great and grievous tribulations suffered the Apostles
  'How great and grievous tribulations did the Apostles suffer?

  (10) Where **doth** the grene knyght holde hym?
  where does the green knight hold him
  'Where does the Green Knight hold him?'

- During the rise of periphrastic *do*, in Early Modern English, the innovatory form is favoured by interrogative over declarative sentences (Kroch, 1989) – a CRE

Periphrastic *do*



Jespersen's Cycle

- English sentence negation exhibits a so-called Jespersen's Cycle: *ne* . . . > *ne* . . . *not* > . . . *not*:

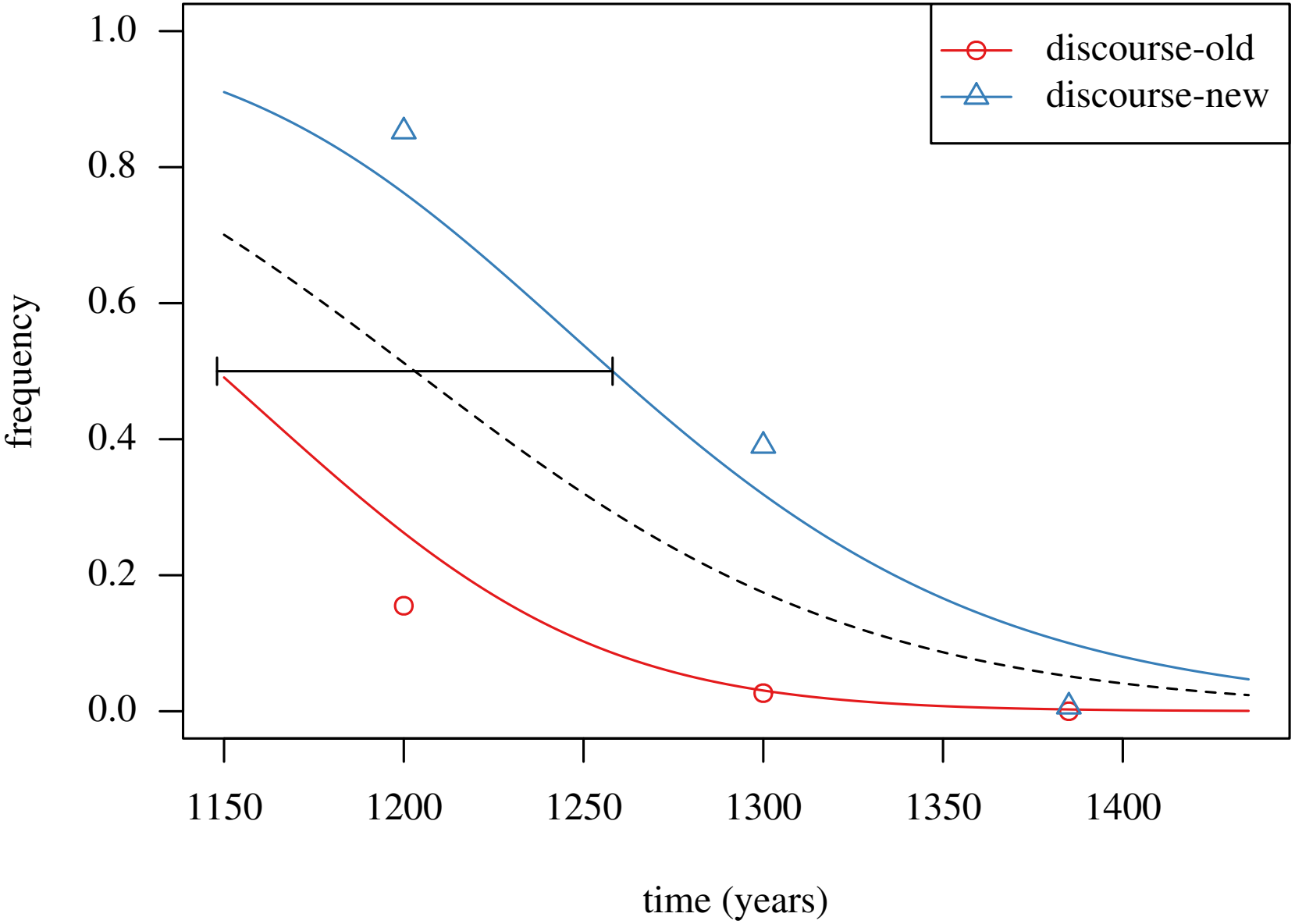
  (11) we **ne** moten halden Moses e lichamlice
  we NEG need observe Moses' law bodily
  'we need not observe Moses' law in body'

  (12) ac of hem **ne** speke ic **noht**
  but of them NEG spoke I not
  'but I did not speak of them'

  (13) I know **nat** the cause
  I know not the cause
  'I do not know the cause'

- Here, we focus on the transition from Stage I *ne* . . . to Stage II *ne* . . . *not* based on data in Wallage (2013). The relevant observation is that discourse-old propositions favour Stage II negation over discourse-new propositions.

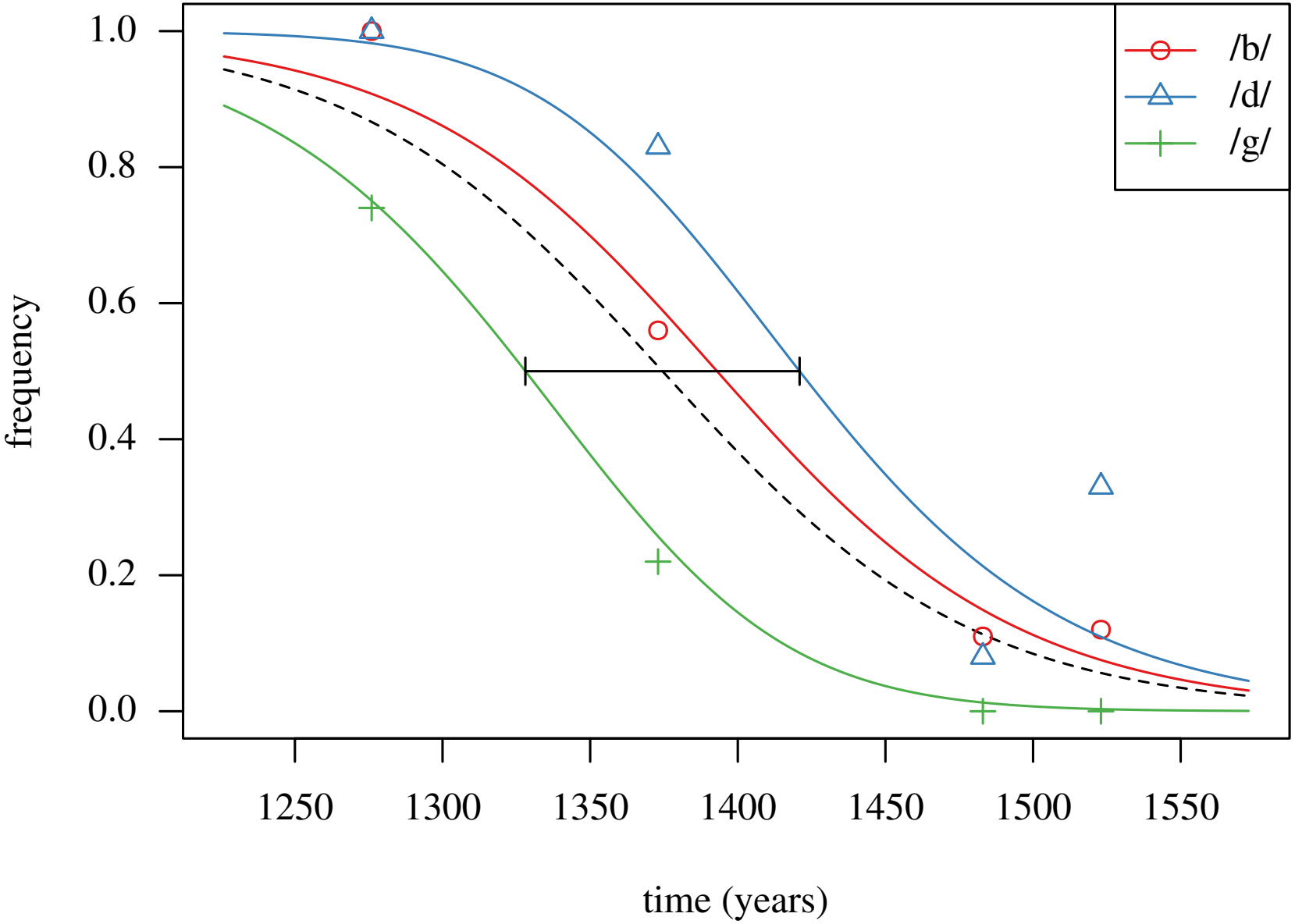
Jespersen's Cycle



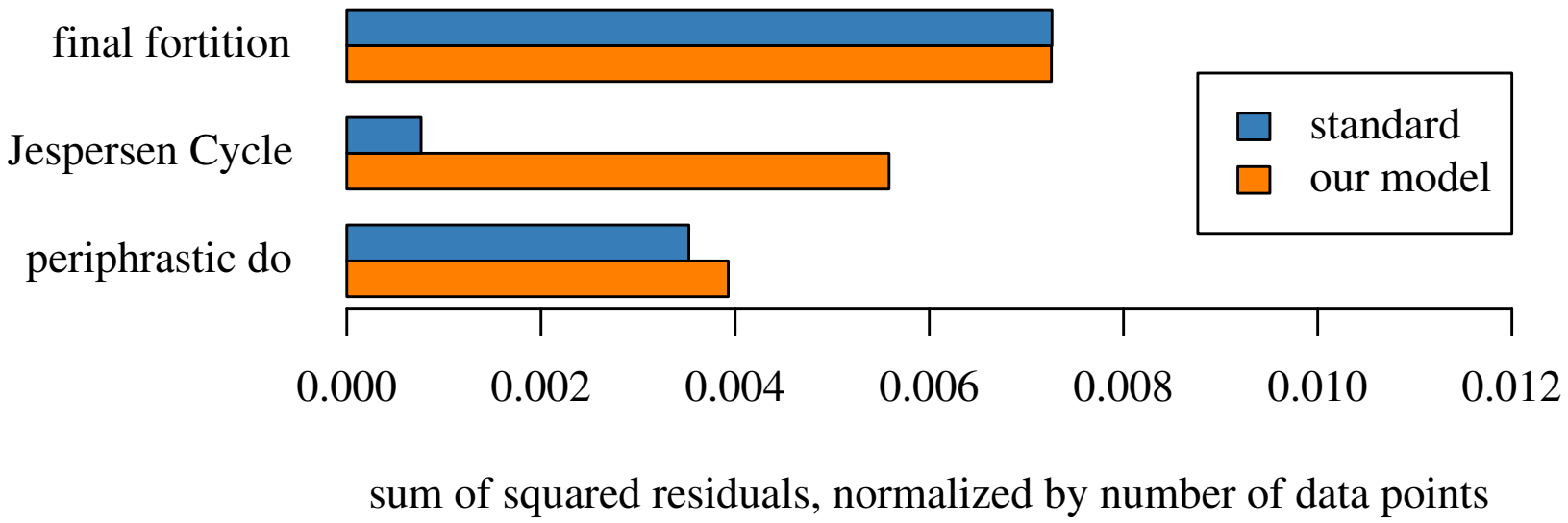
Loss of final fortition

- In Bavarian Early New High German, final consonants begin to be voiced after c. 1000, as testified in orthography (e.g. *tag* vs. *tak*, *rad* vs. *rat*; Fruehwald et al., 2009).
- The CRE: the consonant /g/ exhibits most voicing and the consonant /d/ the least, with /b/ falling in between

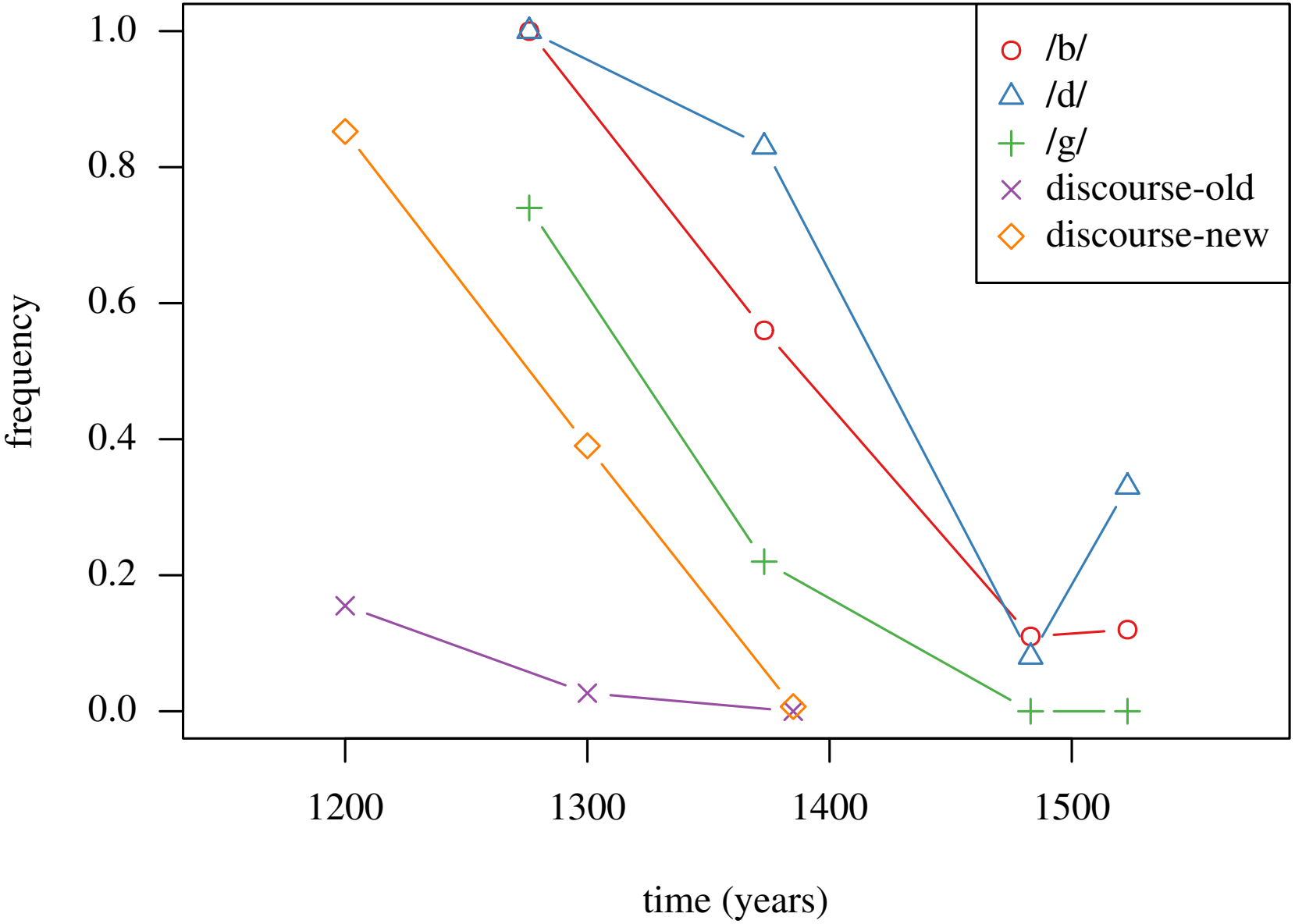
Loss of final fortition



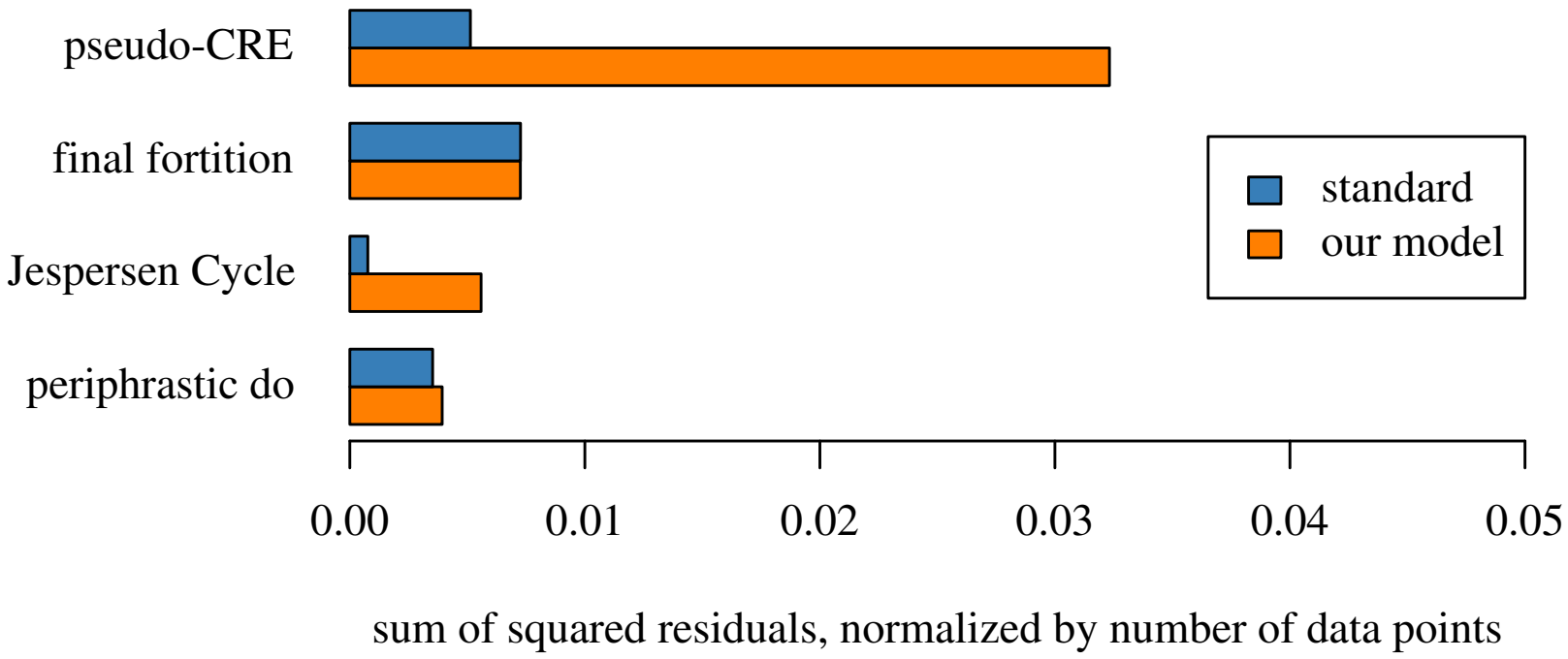
Comparison with standard procedure



A pseudo-CRE


1.0
0.8
0.6
0.4
0.2
0.0
frequency
1200
1300
1400
1500
time (years)
/b/
/d/
/g/
discourse-old
discourse-new

Comparison with standard procedure



Part IV

Conclusion

To conclude

- We have augmented Yang's (2000) model of competing grammars with production biases to account for the CRE
- The model is theoretically motivated and more constrained than the standard operationalization of CREs via sets of independent logistics
- But gives fits to data which are no worse
- And is able to tell (at least some) pseudo-CREs apart from actual ones
- Next steps:
  - Test against more data
  - Solve the central difference equation?
  - Generalize to > 2 grammars
  - Study the model when N is small and language learners are thus unreliable

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